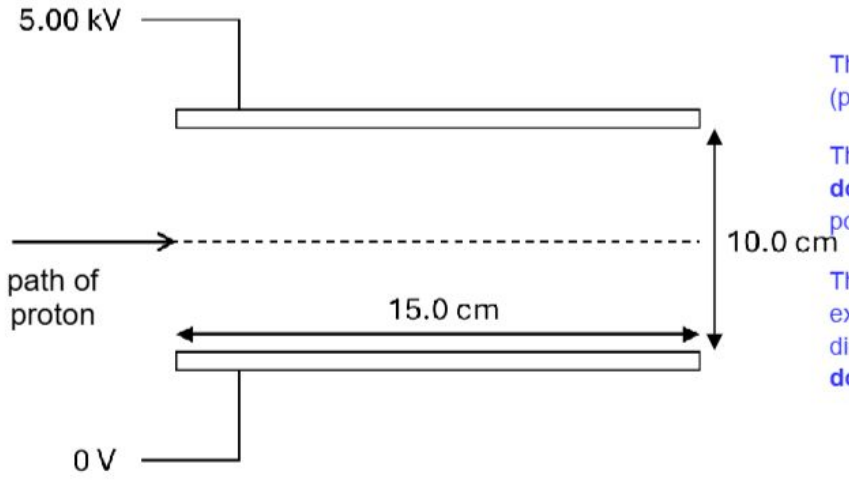
23 A proton, initially travelling horizontally at 2.52×10^6 m s^{-1}, enters a uniform electric field at right angles to the field. The electric field is generated by applying a potential difference of 5.00 kV between two parallel conducting plates 15.0 cm long and 10.0 cm apart.



The top plate is at **5.00 kV** (positive) and the bottom at **0 V**.

The electric field points **downward** (from high to low potential).

The proton (positive charge) experiences a force in the direction of the field — **downwards**.

What is the change in height of the proton as it leaves the plates and the direction which it is deflected?

	change in height / m	direction of deflection
A	8.5×10^{-3}	downwards
B	8.5×10^{-3}	upwards
C	15.6×10^{-3}	downwards
D	15.6×10^{-3}	upwards

$$E = \frac{V}{d} = \frac{5000}{0.100} = 50000 \text{ V m}^{-1}$$

$$F = qE = 1.60 \times 10^{-19} \times 50000 = 8.00 \times 10^{-15} \text{ N}$$

$$a = \frac{F}{m} = \frac{8.00 \times 10^{-15}}{1.67 \times 10^{-27}} = 4.790 \times 10^{12} \text{ m s}^{-2}$$

$$t = \frac{L}{v_x} = \frac{0.150}{2.52 \times 10^6} = 5.952 \times 10^{-8} \text{ s}$$

$$y = \tfrac{1}{2}at^2 = \tfrac{1}{2} \times 4.790 \times 10^{12} \times (5.952 \times 10^{-8})^2 \Rightarrow y = 8.49 \times 10^{-3} \text{ m} \approx 8.5 \times 10^{-3} \text{ m}$$

Ans: A

24 Two point charges each of charge *Q* at a separation of *r* repel each other with a force *F*.